Gods, Games and the Giving Environment: Spirit Beliefs and Cooperation among BaYaka Hunter-Gatherers (Pre-registration)

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1. Introduction

The role that religious beliefs and practices play on social solidarity and cooperation has been a longstanding area of study in the social sciences (Durkheim, 2001; Rappaport, 1999). Moreover, scholars in the quantitative social sciences are increasingly drawing their attention to this relationship, especially among diverse, smaller-scale populations, who have a wider range of religious traditions than are typically examined (Lang et al., 2024; Purzycki et al., 2024a). Despite this attention, little systematic and quantitative work has been conducted among foragers in this area—both in terms of how their religious systems are structured and how beliefs within these systems relate to cooperation. Here, we examine how beliefs in supernatural punishment and knowledge of spirits among the BaYaka foragers of the Republic of the Congo contribute to cooperation within and beyond their local communities, first by mapping some of the local

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religious landscape and then by assessing how specific belief dimensions relate to cooperative behavior.

1.1. Theory

A large literature that draws from disparate theoretical orientations and resides across many disciplines probes the relationship between religion, cooperation, and society (Bulbulia, 2008; Iannaccone, 1995; Malinowski, 1992; Rossano, 2010; Swanson, 1960). One relatively recent focus draws from cultural evolutionary theory and suggests that some aspects of religious beliefs can recruit the psychological motivations that promote cooperation towards other people. More specifically, the "supernatural punishment hypothesis" posits that fear and beliefs in spirits that impose costs on people for misconduct function as effective mechanisms to bolster cooperation by either harnessing prosocial sentiments or curbing anti-social actions (Schloss & Murray, 2011). One account posits that supernatural punishment beliefs are generally widespread because they helped individuals avoid engaging in costly selfish behavior (Johnson, 2015; Johnson & Krüger, 2004). That is, fear of supernatural punishment reduces the likelihood of engaging in acts that are punished by flesh-and-blood agents and what contributes to accounting for the remarkable cooperation expressed by humans is commitment to such deities. Others have expanded this idea to at least partially account for the ubiquity of the so-called "world religions" and their association with social complexity. This view suggests that beliefs in morally-concerned, punitive, and knowledgeable deities should motivate individuals to act cooperatively toward a wider network of anonymous, unrelated individuals who might not ever be capable of reciprocating (Norenzayan, 2013; Norenzayan et al., 2016). This expanded sociality bolsters social complexity insofar as individuals are a part of a more reliable and sustained social order which in turn facilitates success in between-group competition, thus feeding back to more widespread commitment to such gods. This implies that such deities are relatively scarce among small-scale traditions.

Evidence pertaining to these theories is mixed, often relying on different data types and quality, estimands, operational definitions, and assumptions (Purzycki, 2025b; Van Baars, 2023). For example, while some group-level, cross-cultural data suggests that moralistic supernatural punishment beliefs are quite common across society levels (Lightner et al., 2023; Swanson, 1960) and may predate greater levels of social complexity (Watts et al., 2015), others suggest that such beliefs are limited exclusively to very large societies (Roes & Raymond, 2003; Turchin et al., 2023). Furthermore, across a wide range of studies and meta-analyses of various kinds of data, individual-level data from Western samples suggest a complicated portrait of the relationship between religion and cooperation (Galen, 2012; Gomes & McCullough, 2015; Shariff et al., 2016). Despite this inconsistency, individual-level, cross-cultural data suggests a robust association between supernatural punishment beliefs and cooperation toward distant co-religionists, albeit it primarily from "moralistic" or "world" religious traditions (Lang et al., 2019, 2024; Purzycki et al., 2016a, 2024a).

In the case of small-scale traditions of egalitarian immediate-return hunter-gatherer groups, where cooperation and coordination emerge in a dense, face-to-face network, and foundational cultural schemas generally repress authoritative control (Sonoda et al., 2018), expectations around religious patterns are even less clear. Unlike in anonymous societies, the management of social dilemmas in these systems tends to organise via the frequent, direct encounters between members, often resolved in a fully ad hoc manner, and weakening reliance on highly structured, abstract doctrinal systems (Wiessner, 2005, 2014). Indigenous supernatural agents may function through context-specific regulation of social coordination (Boyer, 2021), and even if central, morally concerned figures exist, the punitive nature of control could be in conflict with basic value commitments around autonomy (Gardner, 1991).

Despite the persistent consideration of "small-scale" or "traditional" societies that has driven much of the inquiry concerning religion and society, there remains a dearth of direct, engaged, and systematic studies of foraging societies' religious traditions and cooperation at the individual level or how individual beliefs impact cooperation among these more localized traditions more generally (for exceptions, see Apicella, 2018; Atkinson, 2018; Townsend et al., 2020). In sum, while some variants of the "supernatural punishment hypothesis" suggest that the motivating effect of deities on cooperation should be universal (Johnson & Krüger, 2004), others predict that this effect should be weaker or non-existent in small-scale foraging societies outside such "moralistic" religious traditions (Norenzayan et al., 2016); this is what we aim to assess here among the BaYaka foragers of the Congo Basin.

We are thus faced with two empirical and conceptual challenges: first, the relative absence of quantitative data on religious systems in foraging populations; and second, the limited understanding of how locally salient supernatural beliefs in such small-scale forager populations relate to patterns of cooperation. To this end, we adopt a two-study approach where the first informs the second. We begin by outlining the broader ethnographic and methodological context of our field site and data collection. Study 1 presents a quantitative ethnographic analysis of BaYaka religious life. It aims to map the structure of the religious system by assessing the salience, perceived properties, functions, and relations of local deities on the community level. This provides an empirically grounded account that can be cross-checked with prior ethnographies and used to inform the identification of "moralistic" and "local" deity categories in the BaYaka system that are used within the framework of the broader cross-cultural project. Study 2 then builds on this foundation to assess the relationship between individual-level beliefs—specifically the perception of punishment and omniscience associated with deities—and cooperative behavior toward distant co-religionists, using ethnographic interviews, behavioral economic games, and a causal modeling approach. This design allows us to characterize BaYaka religiosity on its own terms and assess how its specific dimensions relate to cooperation, linking ethnographic depth with quantitative analysis.

2. Ethnographic Background of Field Site

2.1 General Background

In this study, we worked with the Motaba BaYaka, hunter-gatherers belonging to the Western cluster of the Congo Basin forager (CBF) groups (Boyette et al., 2022; Patin & Quintana-Murci, 2018; Quintana-Murci et al., 2008; Verdu et al., 2009). The field site is located near the northern borders of the Republic of the Congo (ROC), in the Likouala Department. As part of the Congo Basin, the world's second-largest continuous tropical forest system, the area comprises dense lowland forests, swamps, and rivers, supporting the highest biodiversity in Africa (Libalah et al., 2026). CBF cultures are separated by thousands of kilometers and speak different languages, but their foundational cultural schemas and institutions remain highly consistent, especially among groups where traditional subsistence profiles have persisted even amid eco-cultural change and state interventions. Central sources of such external influences have been the arrival of settled farming cultures and the subsequent introduction of new dynamics tied to colonialism. Bantu-speaking agricultural groups of the region reached the savanna-forest periphery ~5kya and have been shaping BaYaka lifeways through complex relations. Consistent with the deep-rooted contrast between farmers and foragers, gene flow between the two groups is virtually absent (Bostoen et al., 2015; Boyette et al., 2022; Grollemund et al., 2015). Today, the overall BaYaka population in the ROC is estimated at between 50,000 and 100,000 individuals (Olivero et al., 2016).

2.2 BaYaka life and religiosity

2.2.1 Core Cultural Orientations

As a stereotypical economic disposition of egalitarian hunter-gatherer cultures, BaYaka fall into Woodburn's immediate-return model (Woodburn, 1982), where subsistence strategies yield direct, immediate reward, and consumption occurs shortly after production. This present-orientation is not only apparent in the overall economic stance but exerts its influence across a wide spectrum of emergent cultural elements with specific functions (Bombjaková, 2018; Lewis, 2002; Sonoda et al., 2018). Immediate action, spontaneity, and responsiveness are strongly emphasized over long-horizon planning, abstract reasoning, and pronounced institutional control (Hewlett, 1991, 2014; Lewis, 2002; Sonoda et al., 2018).

BaYaka lack any definite rules of private property. Possession of most goods resembles temporal custodianship; items are distributed according to the present needs of individuals in a demand-sharing system (Bombjaková, 2018; Hewlett, 1991, 1993; Lewis, 2002; Sonoda et al., 2018). Similarly, any attempt to accumulate prestige and gain coercive control over others is met with varied, spontaneously organising counter-practices[2] that employ teasing, rough joking, and

[2] *Mosambo* is a BaYaka 'institution' that organises spontaneously in the camp and is usually initiated by a woman or a group of women. *Mosambo* allows members to express their grievances, opinions, offer advice (sometimes in group level decisions), or refer indirectly to cases of inappropriate behavior (Bombjaková, 2018). Performances can involve multi-modal presentation styles like the bodily re-enactment of events and characters, stylistic speech, and rough, comic critique (Lewis, 2014). Sessions defuse any serious tensions before it can develop into lasting conflict. Even if

avoidance to prevent the consolidation of coercive power and keep authority fluid. Leadership is at most task-specific, and norms strongly discourage the development of durable, generalized political authority. Locally normative value commitments and beliefs are reinforced and reproduced within a highly prosocial environment.

Cooperative production remains essential to women's work and, although conservation regimes and state control led to the predominance of encounter hunting (i.e., opportunistic, direct-pursuit hunting rather than using coordination or trapping), BaYaka subsistence has historically been embedded in a hunting ecology that included the pursuit of both small and large game, where cooperation constituted a structurally central element of the lifeway (Lewis, 2002, 2021; Yasuoka, 2021). Large animals—particularly elephants—held special ritual and economic significance due to both the risks and coordination involved in their pursuit and the quantity of meat they provided for wider sharing networks. (Lewis, 2002, 2021; Yasuoka, 2021). Elephant hunting, even though prohibited since 1997, still has a fully developed ritual-economic frame among the Sangha BaYaka, a close sister group living 100km south of our field site, and even though predation has decreased, frequent encounters with elephants, and the ecological significance of the species sustain its central role in BaYaka cosmology (Lewis, 2002, 2021).

2.2.2 Knowledge and Religion as Embodied Practice

The present-oriented dynamics of BaYaka life significantly shape the modes of knowledge acquisition, transfer, and storage–including that of religious knowledge. Knowledge is rarely externalized as explicit symbolic models or systemized conceptual frameworks. The culture makes little effort to stabilize an objective *body of teachings* through propositional claims independent of bodily engagement. Instead, the known is demonstrated in situated practice or enacted bodily (Bombjaková, 2018; Lewis, 2002, 2014). In teaching practices, thorough verbal explanations are rare, and often framed as intrusive on a child's autonomy or even harmful to their health and sanity (Bombjaková, 2018; Boyette & Hewlett, 2017). Learning is predominantly participation-based and grounded in practice/doing (Boyette & Lew-Levy, 2020; Hewlett, 1991; Lewis, 2002). In line with this, BaYaka religiosity is less a bounded domain of doctrine, characterized by a relative lack of emphasis on formal obligations and the predominance of pragmatic, case-oriented ritual prescriptions with a relatively high individual diversity. Given this weak externalization, everyday practices constitute the primary empirical site where cosmological order is enacted and made visible.

2.2.3 *Ekila*

The concept of *ekila* is central to BaYaka life. Its function unfolds across everyday life and development by anchoring key areas of cosmological knowledge, gender, and political orientations in the physical and biological experiences of human growth (Lewis, 2008, 2016). In everyday

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someone is made fun of during a session, the culture-normative reaction is laughter (Bombjaková, 2018). Collective decisions can also emerge from such social events, like relocating the camp. In these cases, families or individuals who disagree, simply won't follow the group when relocating, often silently, without attracting objection or sanction (Lewis, 2002).

usage, *ekila* appears in a wide variety of contexts that may initially seem unrelated, such as menstruation, blood, taboo, animals' power to harm humans, and various dangers to human reproduction, production, health, and sanity. It is invoked as an explanation for misfortunes like failed hunts and illness—essentially any negative outcomes may be described as having occurred "because of *ekila*." *Ekila* taboos involve prescriptions concerning the timing and conduct of acts. Individuals can "look after" their *ekila*, "mix it around" [with others], or "ruin" it by deviating from proper conduct (Lewis, 2008, p. 298, 307). Universal stages of human maturation (e.g. first period blood) systematically introduce new *ekila* prescriptions and practices, gradually assembling a broader embodied sense for the order of things (Lewis, 2008, 2016; Salali et al., 2019). It is a cultural agent that regulates thought and action, developing ways of sharing food and managing human physical, mental, and reproductive potentials and is understood to benefit the community and individuals alike (Lewis, 2008, 2015). Crucially, *ekila* is not upheld through doctrine or systematic rationalized explanation, but through repeated embodied practice with short verbal additions at most. *Ekila* practices can even differ between two families in the same community (Lewis, 2008).

2.2.4 Forest Cosmology and 'Animism'

BaYaka sociality is further grounded in the constant engagement and emergent intimacy between people and forest ecology. The forest is the direct setting for life and, consequently, an important reference system for thinking about growth, maturation, and social dilemmas. It is a proximate, continuously available model through which local ontological orientations are sustained and reproduced (Bombjaková, 2018; Lewis, 2002). In BaYaka thought, there is little conceptual separation between *nature* and *culture*, or between *natural causality* and *moral order*. BaYaka coordinate the world as one continuous field of relations in which humans, animals, and spirits occupy distinct yet permeable positions. Personhood is not limited to humans but extends to animals, spirits, and other beings 'animated' by everyday direct coexistence–a common theme in 'animistic' cultures[3] (Bird-David, 1999; de Castro, 1998; Willerslev, 2007). Narratives employ a shared vocabulary to describe humans, animals, plants, and spirits alike[4] (Bombjaková, 2018). These are not metaphorical transfers from the social to the natural, but rather reflect a local ontological frame in which primary similarity is grounded in how beings position themselves in the wider eco-moral space.

The world's "giving nature" is expressed in stories about the creator entity, *Komba*, who created the rainforest to be enjoyed and shared by all. In reality, even certain animal species are perceived by BaYaka as failing to align with such egalitarian ideals. In *gano*, fable-like stories,

[3] Even though BaYaka personhood concepts show great fluidity, the system doesn't seem to follow the rules of Amerindian perspectivism. Rare BaYaka folklore includes similar stories of identity shifts between person 'categories' to the ones described by Willerslev (2007) in Yukaghir hunters, who exist in a very similar task-environment and actively use the technology of 'shape/identity-shifts' to locate and manipulate prey species. Köhler (2020) describes convergent beliefs among the Cameroonian Baka, where shapeshifters are called *mokila*, and master-hunters are believed to occasionally shapeshift into elephants.

[4] See: The concept of "*ripening*" and its linked vocabulary in Bombyaková (2018); Animal personhood in Lewis (2002), Chapter 3.

animals are often portrayed as persons whose actions reveal a stance toward the shared world. These often translate observed patterns of conduct into recognizable "types" that can be discussed, mocked, warned against, or used as explanatory templates in stories (Lewis, 2002, 2014).

For instance, the BaYaka contrast their own behavior with that of the *Bilo*, their non-BaYaka neighbors. *Bilo* translates to *uninitiated*, and the term can refer to Bantu agricultural neighbours as well as gorillas (*Gorilla gorilla*). Both are associated with producing a great deal of *motoko* (negative noise; Lewis, 2002) and with being preoccupied with demarcating forest areas for themselves to later defend. This *emic* comparison highlights the BaYaka stance around territoriality, possession, and reactivity. Relational and moral orientation constitute the primary local dimension of likeness rooted in embodied risk and the practical necessity of anticipating how other beings in the forest–human or otherwise–will act.

2.2.5 *Massana*

BaYaka employ a single term, *massana*, broadly translated as "play," to refer to a wide range of activities, united not by a shared literal function but by the joyful, playful mode in which they are ideally exercised. Local proverbs associate the state of *massana* with truth, knowing, learning, maturing, bodily and mental health, and joy. The most prominent ritual event of the culture is called *mokondi massana*, meaning spirit-play, showing how the culture's general emphasis on spontaneity extends to the framing of religious activity (Bombjaková, 2018; Lewis, 2002, 2015).

Mokondi massana is an event in which the camp invites forest spirits (*mokondi*) by creating the right ritual and emotional conditions for their arrival. The group mobilizes rhythm, singing, dancing, and costumes. Even weather conditions and natural surroundings can be used to create the ideal atmosphere; the exact actions depend on the spirit in question, and invitations can fail if the members' synchronization is lacking (Lewis, 2002, 2015). Our informants claimed that dancing with *mokondi* sometimes lasts two or three days, depending on the camp's mood.

Mokondi animate and make palpable the relational dynamics that organize forest life and human sociality. *Massana* is a platform for connecting and negotiating with *mokondi*, building social cohesion in the process (both among group members and between the group and the forest) by harnessing the aesthetic potential of music and collective movement. *Mokondi* can manifest in different people during spirit-plays, but always in initiates, members who have gained access to and actively use the spirit's abilities and guidance. On less frequent occasions, entirely new *mokondi* can be discovered, captured, and brought back to the camp, usually after long, lone forest journeys (Lewis, 2002, 2015).

BaYaka camps also trade *mokondi* among themselves and, importantly, this spirit-economy doesn't follow the normative rules of demand-sharing. For certain valuable spirits, unique to a camp and granting initiates new powers, individuals can pay the worth of four months of labor on occasions (Lewis, 2015, p. 8), though the money received for the spirit is immediately subject to demand-sharing in the community. Spirits can also be exchanged for other valuable items, often for other spirits (Lewis, 2015).

BaYaka informants posit that in *massana*, people heal and acquire important knowledge of ideal worldly relations through experiencing how "beautiful and delicious" spirits are and how joyful and easy they make everyone feel (Bombjaková, 2018, p. 254). During this collective experience, social tensions are recalibrated, and ideal relational alignments are restored. If performed properly, *massana* "casts a spell over the camp" (Lewis, 2015, p. 7) to produce *bisengo*, euphoria-like aesthetic awe for participants to share.

Depending on the invited *mokondi*, certain massanas also have an effect on organising and scheduling hunting and sharing, although with the arrival of novel economic dynamics and an increase in Bantu influence, such effects are less obvious or scaled down to individual- and family-level choices (Kandza et al., 2023). Nevertheless, BaYaka hunters are clear that dancing with *mokondi* renders the forest receptive to human activity (Lewis, 2002, 2015).

It is within this rich and dynamic ecological, cosmological and socio-moral landscape that we aim to, first, describe BaYaka religious life in quantitative terms using the tools of cognitive anthropology and, second, link religious life to cooperation. We first map the structure, salience, and perceived properties of religious entities within the local spirit system, and then assess how beliefs concerning supernatural punishment, knowledge, and moral concern relate to cooperative behavior toward distant co-religionists.

3. General Methods and Sample

Our methods protocols were developed directly from the previous *Evolution of Religion and Morality* Project (ERM; Lang et al. 2024; Purzycki et al. 2016; Purzycki, Henrich, and Norenzayan 2024). We first conducted a preliminary ethnographic Religious Landscape Interview, designed to help us understand the local religious ethnographic context and identify the relevant "moralistic" and "local" gods (Study 1; $n = 20$). After this, we then conducted the core experiments (Study 2; $n = 98$), which consisted of three modules: 1) a brief demographic survey; 2) behavioural experiments to assess cooperation with distant co-religionists; and 3) a religiosity interview specifically asking about two different deities gleaned from the preliminary interview. All materials were first translated into French, then back-translated and edited for clarity and consistency. Yaka—the indigenous language of the BaYaka—is not a written language, and attempts to translate the French questionnaires into Yaka prior to fieldwork proved problematic. As a result, we decided to conduct translations directly in the field during data collection, allowing the questionnaires to be adapted and fine-tuned in context with the help of local translators. All materials in French and the original English are available on the project GitHub page here.

The core protocol had to be adapted to the local context. Although no major changes have been made, initial piloting revealed that questions detached from lived experience or portraying highly hypothetical scenarios could be problematic to many. To reframe such elements in the protocol, with the help of our native translators, we probed alternative wordings with contextual clarifications until a somewhat systematic, working list emerged. Additionally to these, issues with larger quantities also arose; however, in the most important cases—such as age, number of children, and number of co-residents—data were reliably confirmed with the help of local farmers or by asking participants to name their children and co-residents, after which assistants counted and recorded the number of names. We collected these data in the summer of 2025. Participants were all from the Motaba region and grew up in the surrounding camps/villages. It was caterpillar season, so BaYaka were deep in the forest from the early morning, and many went on multi-week journeys to live in forest camps. Before dawn, a team of local translators and research assistants helped recruit participants for both morning and afternoon interview sessions. By the end of our stay, we had saturated the study village and worked with every adult present during the period.

4. Study 1: Spiritual Landscape along the Motaba BaYaka

4.1 Methods

To complement prior ethnographic knowledge and locate appropriate deities for our study, a preliminary ethnographic interview was conducted. To naturalistically solicit deity names among the BaYaka, we used a free-listing task where 20 BaYaka were asked to list as many relevant deities as they could think of, then determine whether each: i) was concerned with what people do or how they behave (moral concern); ii) could see into peoples' minds or know their thoughts or feelings (omniscience); iii) punish people who behave in ways the deity does not like (punishment); and iv) reward people who behave in ways the deity likes (reward). These were all binary "Yes/No" questions. Free-listed deities were later also ranked based on how powerful they were to participants.

Free-lists offer the advantage of capturing culturally embedded knowledge as it is naturally organized and recalled by participants, thereby preserving ecological validity (Purzycki, 2025a). To assess cultural salience of BaYaka deities, we calculated Smith's *S* (Smith & Borgatti, 1997) of each listed spirit, which is defined as follows:

$$S = \frac{\Sigma(\frac{n + 1 - k_i}{n})}{N}$$

where k_i is the numeric order in which an item i appears in an informant's list, n is the list length, and N is the sample size. As such, if every participant listed a particular deity *and* listed it first, *S* = 1.00. To propagate uncertainty around deities' cultural salience scores, we modelled each item type independently in ordered beta-distributed models with uninformative priors (Major-Smith & Purzycki, 2026). We use the AnthroTools package (Purzycki & Jamieson-Lane, 2017) for R to perform all analyses.

4.2 Results and Discussion

In total, the BaYaka listed 20 different deities, with the cultural salience of the top-eight deities displayed in Figure 1, with perceived "power" rankings in Figure 2. Punishment, knowledge breadth, moral concern and reward profiles of the 5 most salient entities are shown in Figure 3.

For reasons we now discuss, we selected *Komba* (a creator deity) as our moralistic deity, and *mokondi* (forest spirits) as our local deity.

As the forest's creator and the figure from whom sharing rules are derived, *Komba* was our primary candidate for the moralistic deity role prior to the study (Lewis, 2002, 2008). *Komba* placed only second to *Ngoku* on the power ranking list (Fig. 2). *Ngoku* was a gender-specific spirit, exclusively reported by women, with a significantly lower salience score due to only-female relevance (Fig. 1). Although *Edjengi* and *Bolobe*—two spirits whose *massana* are the most frequently practiced at the site—placed higher on the salience list, in the follow-up questions, *Komba* was uniquely attributed with omniscience by most (the highest score among entities), and all informants perceived him as rewarding and morally concerned, making *Komba* the only character to reach the highest scores in the combination of these domains (Fig. 3). *Komba's* centrality also shows in a balanced female-to-male listing ratio, indicating gender-independent importance (Fig. 1).

The remainder of the most salient spirits listed all fall under the aforementioned umbrella of *mokondi*. *Mokondi* are more specialized entities than *Komba*. They are clearly divisible by gender-based piety and are tied to practices and characteristics culturally associated with gendered roles, which accounts for the more peripheral position of certain practice- and/or gender-bound *mokondi*. One gender-bound example is *Ngoku*: the most salient and powerful female spirit, and also the most punitively perceived deity (Fig. 3), about whom women informants were highly secretive during open-ended follow-up questions, and whom male participants did not list at all (Figs. 1 & 2). This suggests a highly specific profile and function of the spirit, unfolding within female domains of the local lifeway—yet within that domain, serving a more central role than any other deity (Fig. 2). *Elimu*, *Eloko*, *Mayobe*, and *Soh* are examples of less salient *mokondi*, which show no co-mentions with one another, emphasizing their importance for specific individuals, but indicating no significance on the community level (Fig. A1 in Appendix). Follow-up questions revealed these unique, individualised relations between these *mokondi* and hunters who listed them, alongside their complete irrelevance for those who did not.

Edjengi, a spirit to which all adult men at our field site were initiated and the most frequently invoked *mokondi*, emerged as our primary candidate for the site's local deity. *Edjengi's* cosmological and practical importance was clear from open-ended questions about the spirit and living with the community. This consensual cultural importance is clearly reflected in the free-list data; despite being a male spirit, his listings were relatively balanced across genders (Figs. 1 & 2). He has been described as "the most powerful [water] spirit," the only spirit who can "kill without consequences," and the "first husband" of all women who has been captured by BaYaka men collectively to establish female marital autonomy and human society (a version of the story in Lewis, 2002). According to our informants, *Edjengi's* powers are still essential to male-oriented forest use; the spirit is associated with courage, foresight of danger, and vitality.

Bolobe's profile shows high resemblance to *Edjengi* with similar centrality (Fig A1), and a somewhat higher male bias and lower scores in cultural salience and perceived power (Figs. 1 & 2). *Bolobe* is a central "abundance" *mokondi* whose *massana* provides hunters with important visions of the subsequent hunt and later dreams of the "prey's blood" and location (information from our participants, but also found in Lewis, 2002, 2015). Consistent with these stories, *Bolobe* was perceived as highly rewarding, second only to *Mipudza*—a far less salient but functionally similar male-dominated abundance-type hunting mokondi (Figs. 1 & 3). The early morning sight of *Téndé* (Cinniris sp.), *Komba's* songbird messenger in the creation myth, is occasionally mentioned by hunters as further proof of a successful *Bolobe massana*. *Téndé's* presence in the morning means that *Bolobe* "opened the camp" (Lewis, 2015, p. 13), and the forest is now receptive to hunting and foraging activity. In the hunters' narrations (also in Lewis, 2002, 2015), *Bolobe* can fly over the canopy to gather prey; in other versions, he lives with *ngulu* (*Potamochoerus porcus*), sleeps under tall trees, he has translucent skin, a large belly, and a thick beard and dresses in fresh, green leaves when visiting the camp. Tsuru lists similar descriptions of *mé* (forest-spirit) among the Cameroonian Baka (a close sister-culture), such as a "big head", "white skin", "long beard", and "bulgy eyes" (1998, p. 60). As a relevant addition, *Bolobe* was already documented in the late 1990s ethnographies of the Mbendjele by Lewis (2002, 2015), where he is referred to as *Malobe* and performs a similar function. Lewis (2002) also noted the spirit's absence among the Motaba BaYaka (our focus group) at the time, but since then, *Bolobe* has spread through processes of spirit exchange and now ranks as the second most salient deity in our study, demonstrating the highly flexible nature of the *mokondi* system.

The *mokondi* system is a partially shared, partially individualized set of relations that are repeatedly mobilized to guide action—this repertoire can flexibly be utilized to draw on, assemble, and stabilize individual spirit-profiles based on individual preferences and context. While the category of *"mokondi"* (spirit) itself is a culturally stable concept among the BaYaka, participants individually fashioned strong personal relations with otherwise peripheral spirits, or reframed the culturally foregrounded functions of more salient spirits in light of personal experience and needs. *Mokondi* are clearly perceived as more directly accessible agents than *Komba*. According to informants, they inhabit forest spaces, hunt on their own, often form camps in the forest, "herd" animals, get involved in disputes, and maintain personal, dynamic, evolving relationships with human individuals, just as persons do within social groups. Open-ended questions and subsequent in-depth conversations with hunters revealed that, at times, *mokondi* generate bodily sensations that signal their presence, particularly during massana and forest trips. Informants frequently described actual reciprocal forms of sharing between the human community and *mokondi*, including leaving hunted animal carcasses in the forest for Edjengi and Bolobe to sustain good relations.

Largely due to their general relevance, fluidity, relative lack of omniscience, and the individualistic and often gendered nature of specific *mokondi* spirits, we decided to use the umbrella term *"mokondi"* to refer to forest spirits in general/collectively, allowing participants to draw on their own associations when responding questions concerning the local deity.

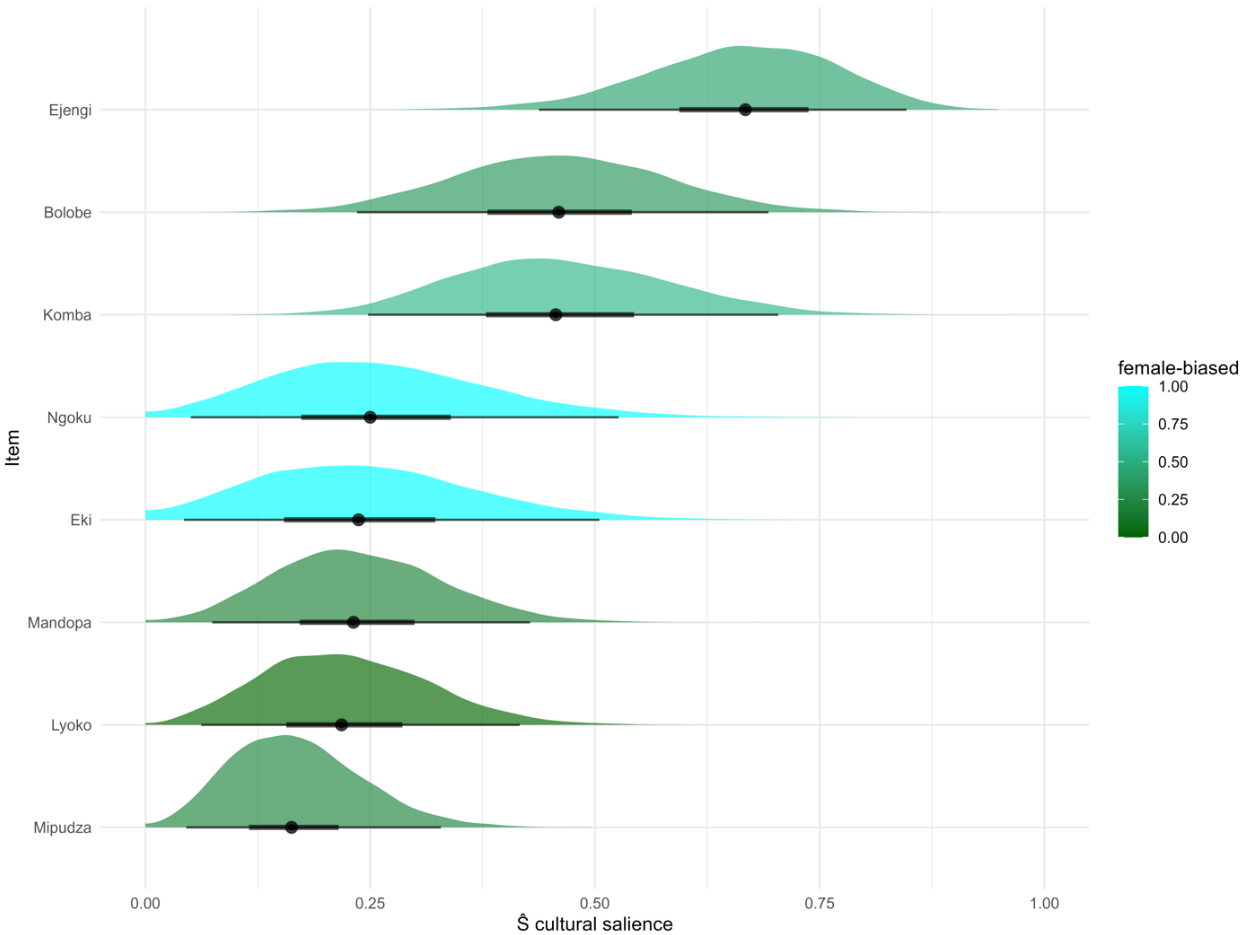

Figure 1. Cultural model of BaYaka deities (*n* = 20). Data are based on the frequency and order in which BaYaka deities were free-listed. Connection values are Smith's *S* cultural salience, with 50% credible intervals in thick lines and 95% credible intervals in thin lines. A value of "1" means that all participants listed said deity and listed it first. The colour gradient shows the male-to-female ratio of free-list mentions, adjusted for differences in sample sizes (1 = female; 0 = male).

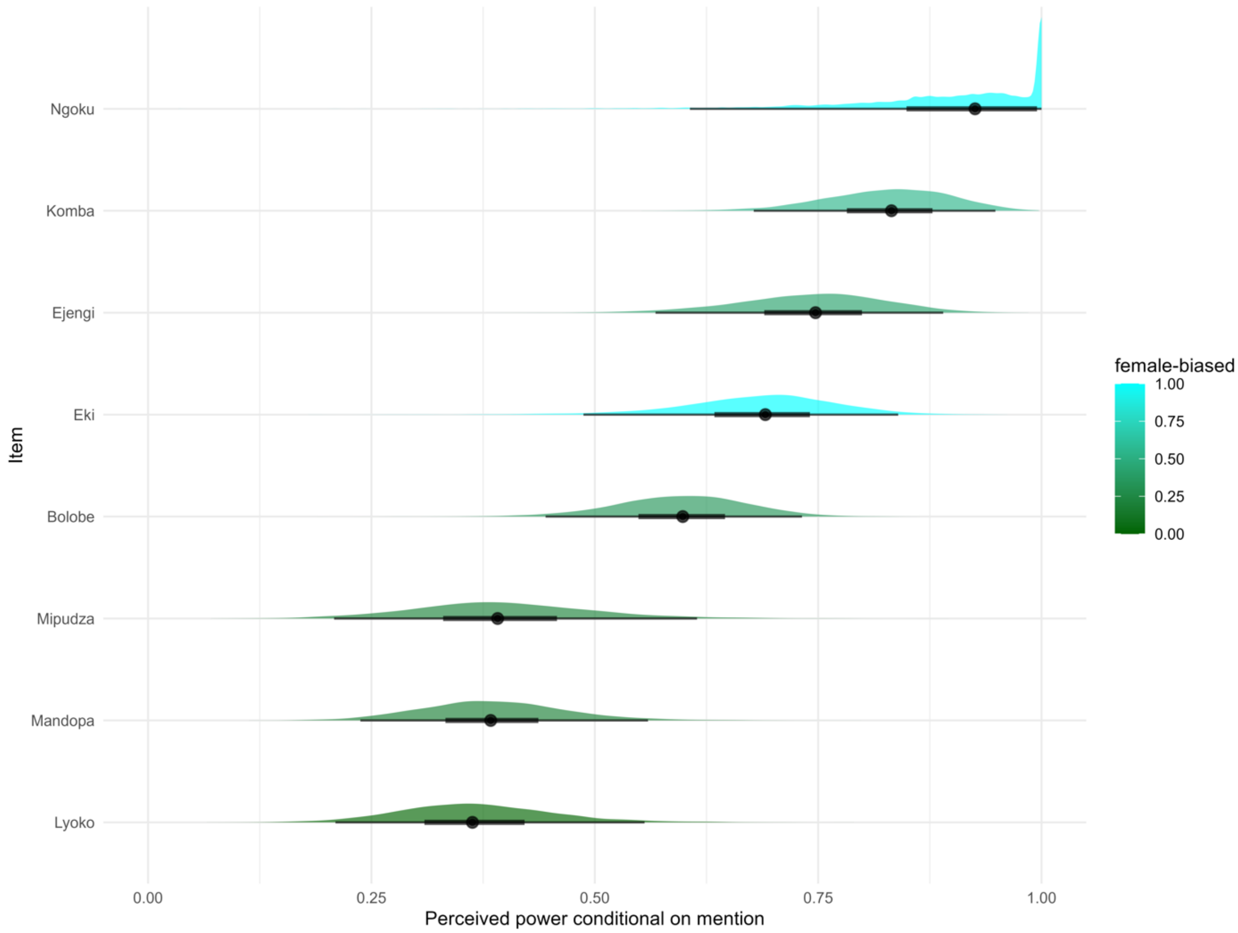

Figure 2. Perceived "power" of BaYaka deities conditional on mention (*n* = 20). Connection values are rank-based salience scores derived from deity power/importance rankings, calculated only among respondents who mentioned a given deity. Thick lines indicate 50% credible intervals and thin lines indicate 95% credible intervals. The colour gradient shows the male-to-female ratio of mentions, adjusted for differences in sample sizes (1 = female; 0 = male). Note the high density of values at "1" for Ngoku—as most participants said this deity was the most powerful, when nominated—which "squashes" the distributions for the other deities.

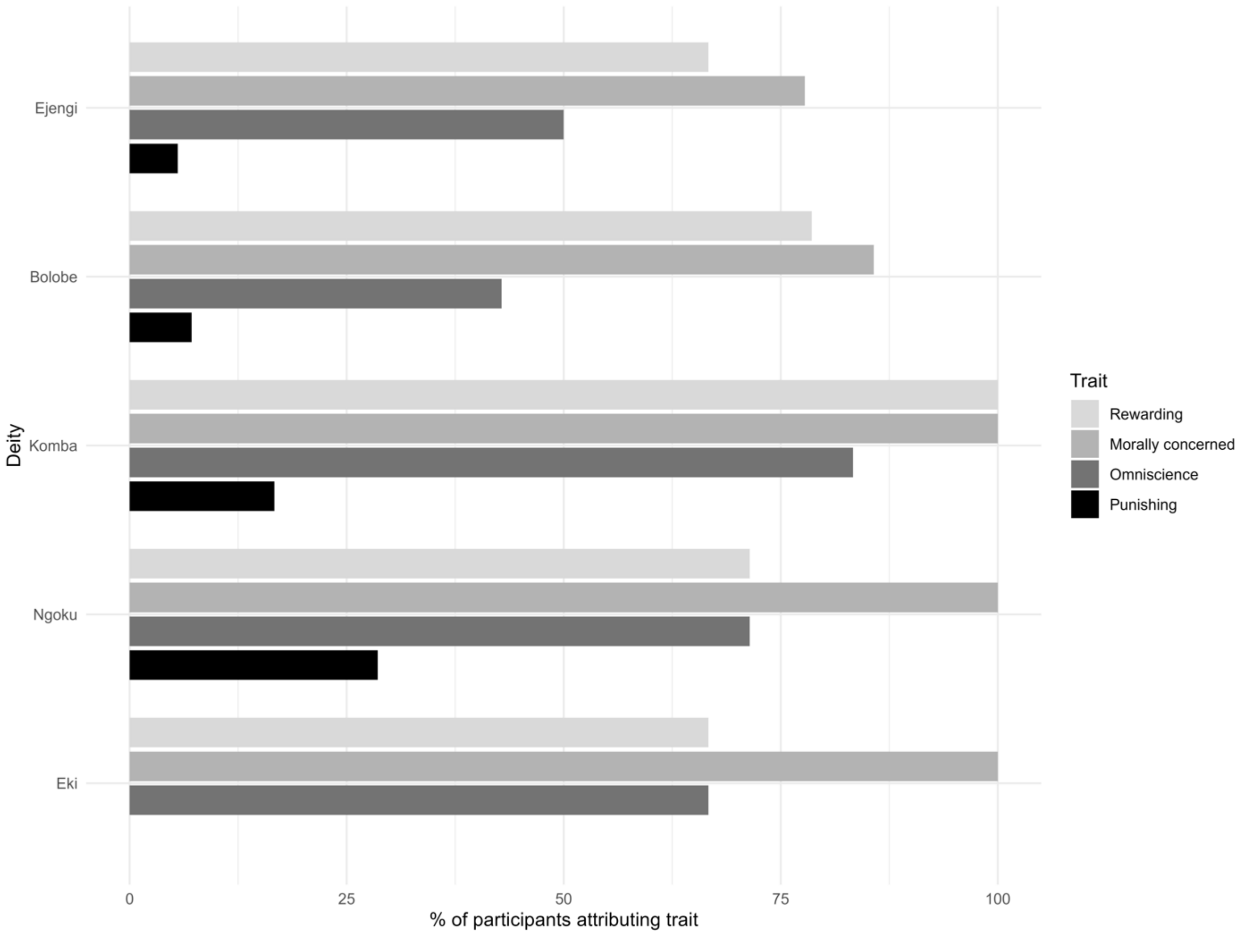

Figure 3. The five most salient BaYaka deities and their perceived traits (*n* = 20).

5. Study 2: *Mokondi* beliefs and cooperation

5.1 Methods

5.1.1. Theoretical Model

For this second study we explicitly take a "causal inference" approach to the question of religious beliefs and the expansion of cooperation (Hernán & Robins, 2020; McElreath, 2020; Pearl et al., 2016; Rohrer, 2018). That is, we have a clear *estimand* in mind – i.e., the target quantity we want to estimate (Lundberg et al., 2021) – which here is: *the joint causal effect of supernatural punishment and omniscience beliefs towards local deities (here, forest spirits) on cooperation with distant co-religionists of the same local religious tradition* (this will be unpacked in more detail below).

Based on this estimand, we then have to detail our assumptions necessary to estimate this causal effect without bias. A key consideration is that of *exchangeability*, meaning that there are no open confounding paths between our exposure (supernatural beliefs) and our outcome (cooperation) resulting from any common causes of these variables. That is, we have to adjust for the potential biasing effects of any confounders in our statistical model. Importantly, in addition to adjusting for confounders, we need to avoid *overadjustment bias*, by not including mediators—variables caused by the exposure that in turn cause the outcome—and colliders—variables caused by both the exposure and outcome—in our models (McElreath, 2020; van Zwieten et al., 2022). The inclusion of variables in our statistical model therefore has to be based on causal considerations, which importantly requires *extra-statistical* assumptions, as causality cannot be inferred from data alone (D'Agostino McGowan et al., 2024).

These causal assumptions can be encoded in Directed Acyclic Graphs (DAGs; Hernán & Robins, 2020; Pearl et al., 2016). Our causal assumptions for this study are summarised in the DAG in Figure 4 (this model was adapted from our core causal model for the wider project, available here). Here, we assume the following variables are core confounders of our estimand: age, sex, education, number of children, material security, and early socialisation/exposure to non-BaYaka lifeways. These assumptions are based on ethnographic expertise, previous literature suggesting that these factors plausibly impact both religious beliefs and cooperation (Baimel et al., 2022; Purzycki & Bendixen, 2024; Schwadel, 2015; Vardy et al., 2022), as well as logical considerations (e.g., nothing can cause age or sex, meaning they can only be confounders, while education is largely fixed earlier in life, hence is more likely to be a cause of these variables rather than a consequence). Based on ethnographic knowledge, early socialisation and integration with non-BaYaka communities, institutions and markets are also plausible confounders as they likely impact patterns of cooperation and beliefs regarding local forest spirits. For instance, even at young ages cooperation among BaYaka children is much fairer than among neighbouring Bantu children (Boyette & Lew-Levy, 2019; Hewlett, 2017; Lew-Levy et al., 2026), while exposure to towns and markets may impact sharing norms and dynamics (e.g., town living has been associated with less temporal discounting among Mbendjele BaYaka; Salali & Migliano, 2015). While we lack the fine-grained data necessary to assess this in detail, we do possess information on whether they have ever lived in a village/city, and which village they currently reside in (one being closer to a logging town, and hence more market integrated), which may proxy this to some extent ("years of formal education" is also likely to proxy this to some extent). Although we believe these assumptions to be plausible, given these cross-sectional data we cannot rule out alternative biasing paths. For instance, we are assuming that education, number of children and material security are *causes of*—rather than *caused by*—religious beliefs (i.e., they are confounders, rather than mediators), as well as assuming that cooperative behaviour does not cause religious beliefs (i.e., there is no reverse causality between our exposure and outcome).

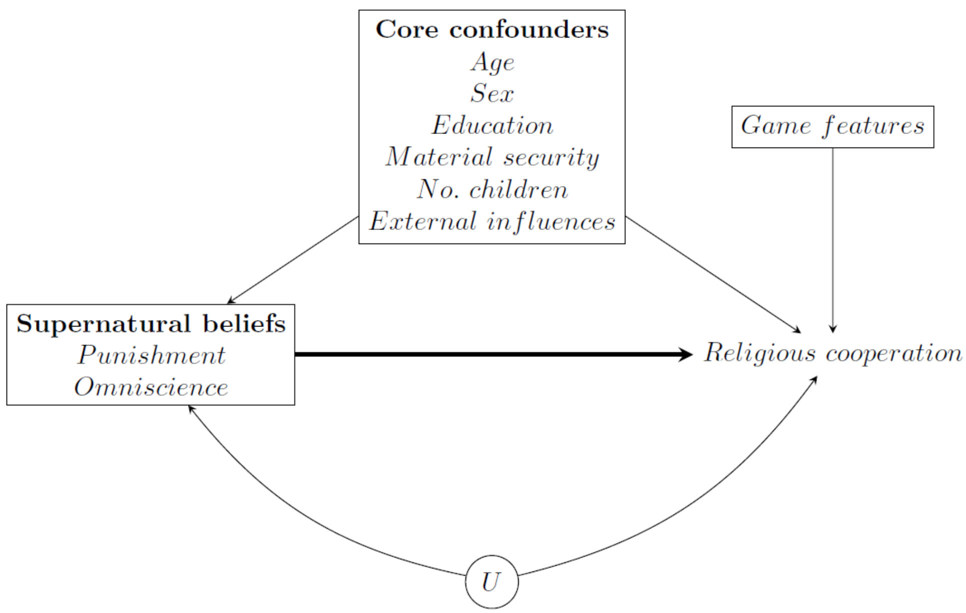

Figure 4: Basic Directed Acyclic Graph (DAG; see text for description). The bold arrow denotes our estimand of interest. Note that causal relations between the core confounders are not displayed here. Causal relations between the components of supernatural beliefs are also not displayed here, as our estimand is the joint effect of supernatural beliefs on religious cooperation (see the main text for more discussion on this).

Some additional features of this DAG and further assumptions required for a causal interpretation are worth noting. First, there is a node "*U*" to denote potential unmeasured confounding. While we endeavoured to include all relevant confounders, this is nonetheless an assumption and we cannot rule out additional unmeasured confounders biasing our estimates. For instance, while we use "ever lived in a village/city" and "current village" as proxies for external non-BaYaka influences, these are not direct measures and so are unlikely to fully capture this influence, meaning there is potential residual and unmeasured confounding. Second, the DAG includes a node for "game features," which includes factors such as the order in which games were played and the number of comprehension checks needed. These are not technically confounders—as they may only cause cooperation, not religious beliefs—so their inclusion will not reduce bias, but may help improve precision in our estimates. Finally, for a causal interpretation to hold we also have to assume that there is no selection bias (i.e., that our sample is representative of the target population; here, wider BaYaka) and no measurement bias (i.e., that all variables are measured without error).

These causal assumptions are all embedded in our theoretical causal model (DAG in Figure 4), which feeds into our statistical modelling approach discussed below. We will return to whether we believe these assumptions are met, and potential implications for interpreting our results, in the Discussion section of the final paper.

5.1.2. Procedure

As briefly described above, our experimental protocols consisted of three modules: (1) demographics interview; (2) a religiosity interview about *Komba* and *mokondi*; and (3) the behavioral experiment. We detail each below, along with explicating any necessary deviations in the protocol.

5.1.2.1. Demographics

We asked a range of questions about sociodemographic factors and food/material insecurity. Of particular importance here are variables we treat as confounding factors, namely: age, sex, number of children, total years of formal education, food/material insecurity (an average of four binary questions asking about whether they are worried about having enough food in the next month/6 months/year/5 years), and two proxies for external non-BaYaka influence (whether they have ever lived in a village/city, and current camp).

5.1.2.2. Religiosity variables

Our primary set of exposure variables were stated beliefs about forest-spirits' (*mokondi*) punishment and breadth of knowledge (referred to here for brevity as "omniscience"). For the former, we used the mean of two binary questions (yes = 1, no = 0): (1) Do forest-spirits ever punish people for their behavior? and (2) Can forest-spirits influence what happens to people after they die? Similarly, we measured forest-spirits' attributed omniscience with the mean of two questions: (1) Can forest-spirits see into people's hearts or know their thoughts and feelings? and (2) Can forest-spirits see what people are doing if they are far away, in Brazzaville?

5.1.2.3. Behavioral experiments

We operationalized cooperation using two variants of two different behavioral economic experiments. To measure impartial rule-following (i.e., fairness and honesty), we used the Random Allocation Game (RAG; Hruschka et al., 2014; Jiang, 2013). To measure generosity, we used the Dictator Game (DG). All participants played the Random Allocation Game first. Each game type consisted of two, randomly ordered, variants. In one variant, participants allocated tokens between an anonymous co-religionist local community member vs an anonymous, geographically distant co-religionist. In the other variant, participants allocated tokens between themselves vs *another* anonymous, geographically distant co-religionist. Co-religionists were explicitly defined as BaYaka spirit-initiates from either their own camp (for the local recipients) or from a distant BaYaka camp (for the distant recipients).

For both games, allocations were put into opaque cups, labelled with each recipient. Participants used blank wooden tokens, each worth 30 XAF (Central African CFA francs; approx. $0.05 USD). In the Random Allocation Game, participants played each variant with 30 tokens. In the original RAG protocols, participants were asked to think of which cup to put each token in and then roll a fair two-coloured die; if the die came up black, participants were supposed to put the token in the cup they were thinking of, else put the token in the other cup if the die was white. Participants played anonymously by themselves, meaning they had the opportunity to bias allocations towards one of the recipients and away from the other (i.e., away from the 50/50 chance per roll). Comprehension checks were used to ensure that participants understood the rules of the RAGs. In the DG, participants played each variant with 10 tokens, which they simply divided between each of the two recipients as they wished. In addition to their winnings in games, each individual was paid a participation fee of 400 XAF.

This original version requires participants to mentally select an option, retain that choice, and then execute a rule based on a die outcome. In preliminary trials, this proved challenging for a great portion of BaYaka participants, as such operations involving newly introduced abstract symbols and multi-step rule-following are relatively rare in the everyday forager task environment, which is typically grounded in direct, embodied interactions, with limited reliance on symbolic combinatorial operations. Instead, to culturally frame the game, participants were asked to place their hand on the cup of their choice—thereby physically marking their decision—before rolling a die with a hand image (a more direct marker of the hand-choice rather than a color) on three sides and nothing on the other three sides. If the die showed the hand symbol, they placed the token in the cup they were touching, otherwise, in the other cup. The use of a hand symbol (rather than abstract colors) and the physical marking of choice made the task more intuitive.

While this adaptation introduces a potential concern that choices might become more observable compared to the purely mental protocol (Hruschka et al., 2014), games were played in full privacy, and the research team made a serious effort to treat payouts as anonymously as possible. Moreover, post-game conversations suggest that participants clearly understood the complete privacy of their decisions. Thus, we believe these adaptations improved participants' comprehension and engagement with the task without undermining the game's core logic, and increased the ecological validity of the measures.

5.1.3. Statistical model

Given the causal model outlined above (Figure 4), our statistical model flows directly from these assumptions. Our outcomes are the number of tokens allocated to distant co-religionists, in each of the games and conditions described above (self vs distant and local vs distant conditions, for both the RAGs and DGs). Our exposure is the joint effect of forest spirit punishment and omniscience beliefs. We focus on the *joint* effect here due to the difficulties of knowing the causal relations between these variables (i.e., do punishment beliefs cause omniscience beliefs, or vice versa, or reciprocal causation between both?). This allows us to side-step these complex issues, albeit at the expense of not being able to estimate the independent effects of both. Our confounder set includes the putative confounders identified above: age, sex, education, number of children, material security, ever lived in a village/city, and current camp of residence.

Our regression models will be aggregated binomial models for the RAG outcomes, and ordinal models for the DG outcomes. All models will include the forest spirit punishment and omniscience mini-scales as exposures, along with the above confounders and game features as covariates. From conducting the experiments, entering the data, and performing audit checks on the data, we observed a preponderance of hyper-fair behaviours in the RAGs among the BaYaka. That is, in the Self RAG condition more participants gave 15 tokens than expected under a theoretical binomial distribution if participants were playing the games fairly; similarly, in the local RAG condition participants gave more 15, 16 and 17 allocations to distant recipients than would be expected (potential reasons and implications of this will be discussed in the Discussion section of the final paper). As game behaviour may therefore be a mixture of strategies—i.e., some participants allocated 15 (or 16, or 17) tokens regardless of the die, while others followed the rules of the game more faithfully—assuming a single aggregate binomial distribution to model these behaviours may be erroneous. Instead, bespoke "inflated" binomial models will be used for these RAG results; a "15-inflated" binomial model for the self RAG, and a "15/16/17-inflated" binomial model for the local RAG. These mixture-models first model the probability of an excess value from the underlying distribution (i.e., a logistic regression model), followed by an aggregated binomial model for all other values. Given the relatively small number of excess values (approximately 10 to 20, depending on condition) and the large number of potential covariates, we will not include covariates in the "inflated" part of these models, and instead simply model the rate of excess values.

To estimate the joint effect of our exposures we will use a *g*-computation approach (Hernán & Robins, 2020), which compares the marginal counter-factual states when both exposures are coded as "0" (i.e., no belief in forest-spirit punishment or omniscience) to both coded as "1" (i.e., belief that forest-spirits both punish and are omniscient), based on posterior predictions from the regression models. In addition to using formal *g*-computation causal methods, this approach also translates model outputs from difficult-to-interpret statistics (i.e., log-odds, or odds ratios) to a more natural and easily-interpreted scale (i.e., difference in number of tokens), especially given our use of "inflated" mixture models, which complicates interpretations based on regression coefficients alone. Additionally, to test whether the relationship between punishment and omniscience on cooperation is linear or multiplicative, we will compare whether the inclusion of an interaction term between exposures improves model fit (using LOOIC model comparison statistics).

All models will be conducted in the R programming language (R Core Team, 2024), with Bayesian models using Stan (Stan Development Team, 2026). All models will be run with four chains with 2,000 iterations (1,000 as warm-up), with visual inspection of mixing between chains, r-hat values close to 1.00, and sufficient effective sample sizes used to ensure the chains had converged and were sufficiently-powered. Weakly-regularising priors will be used throughout.

For a demonstration of our analytic approach, see here.

5.1.4. Pre-Registration

Data for this study has already been collected, cleaned and processed ready for analysis. The methods protocol, however, was designed prior to data collection as part of the project's grant proposal, available here. When preparing this pre-registration we therefore had some information about the data, which was useful to help understand the specific ethnographic context (e.g., the broader religious landscape), and any complexities of the data (e.g., excess 15 [and other] values in the RAG data, any necessary changes to protocols, etc.), which together helped inform both our causal model and statistical analyses (e.g., use of "inflated" aggregate binomial models for the RAGs, choice of confounders). Nonetheless, while we have had access to this data, we have not conducted any preliminary analyses on these data, so do not know the relationship between the exposures and outcomes.[5]

[5] Note that Study 1 was not pre-registered, as it was designed to map the local religious landscape and inform the selection of moral and local deities for the subsequent post-game religiosity interviews. As such, it serves as a descriptive and preparatory/exploratory step rather than a hypothesis-testing component.

Disclosure statement

The authors declare no conflicts of interest.

Authors' roles

MIK collected the data with the help of MMCE, DGB, YRO, and KAA, AHB and VKH facilitated the field work, DM-S and BGP managed the project and contributed to analysis, and BGP contributed to designing the protocols. MIK, DM-S, AHB and BGP contributed to writing this manuscript.

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Data availability statement

Data and analysis code for Study 1 is available on the project GitHub website, while data and code for Study 2 will be added once the pre-registered analyses have been conducted: https://github.com/bgpurzycki/GGSL-Project/tree/main/Code/Objective%201/BaYaka.

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Appendices

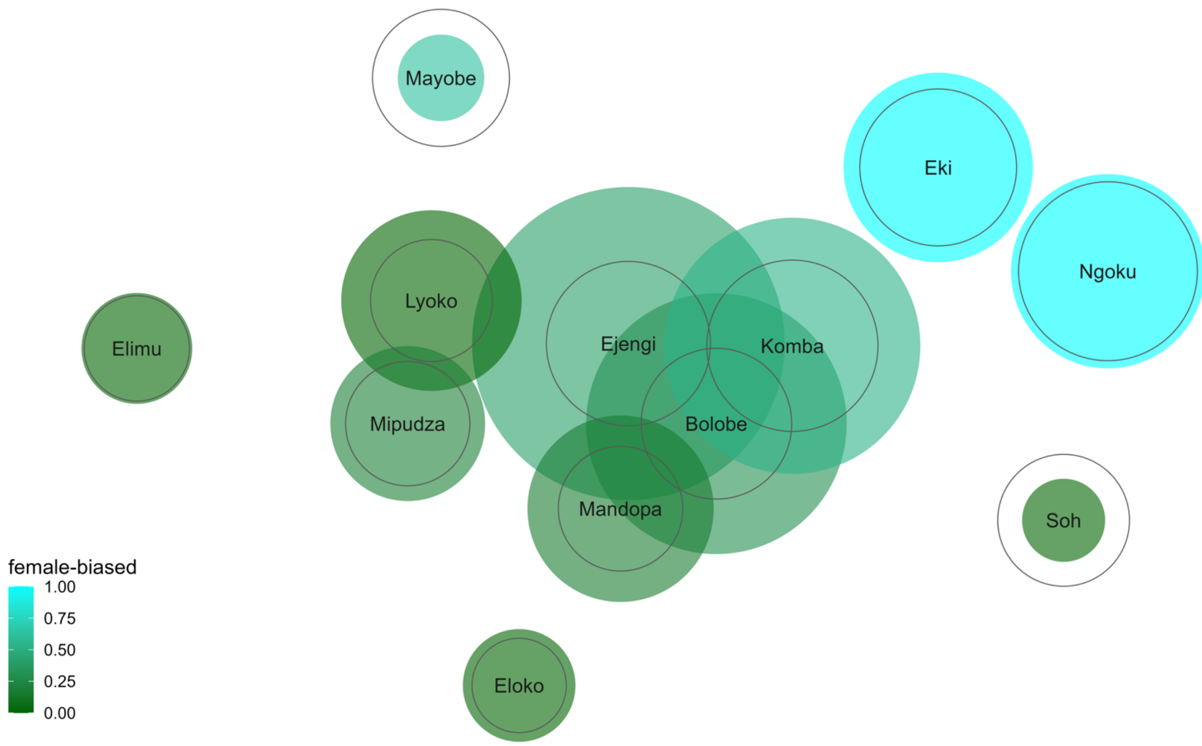

Figure A1. Semantic association network of local religious entities (*n* = 20). Colored node size is proportional to each deity's cultural salience (Smith's *S*), while empty circles represent perceived power conditional on mention. Node color reflects sex-balanced mention proportions, ranging from female-biased (turquoise) to male-biased (dark green). Node positions are derived from weighted co-mentions, such that deities mentioned together more frequently are positioned closer to one another.